PREPARED BY ADIL ALVI

MTH202 MIDTERM SUBJECTIVE QUESTIONS 2019 – 2010

1. Given a diagram find out whether the given 3 arguments are right or wrong
2. Normalize a given table to 3rd degree
3. Draw a cross reference matrix table
4. What 2 considerations are performed by operating system when
5. How many term are there -9-6-3+..... To 66...
6. Member ship table tha conditions k mutabiq bnana tha ..
7. Domain range wala tha..
8. Conditions k muutabiq connectives lgane they

1) Define arguments
2) Show axa is similar to equivalency relationship.
3) Find real valued functions for given polynomial.
4) Give power set of following set...(x, y)
5) Find f (2) while f (0) was given and following relation also provide.

1) Truth table ik 5 marks ka tha
2) ik squence series wala aya tha
3) ik graph bnana tha relation ka .
4) Fog(x) waly 2 question thy

Question 1: if -9, -6, 3, 66 series then find its nth term.
Question 2: f(x) = 2x + 1 and g(x) = x2 – 1
Question 3: (p∨q) ∧ ~ (p∧q) truth table
Dec 14, 2019

1) Truth table tha
2) fog(x) gof(x) sa zidah ppr tha
3) define argument
4) Q tha js me batna tha kon sa law ha (implication negation)
Dec 17, 2019

1) if f and g are one to one function then prove that fog is also one to one function (5 marks)
2) shade the given venn diagrams (5 marks)
3) if a is a set of vowels and b is set of consonants then prove is this true or not. That an intersection b = b intersection a (3 marks)
4) construct the truth table
Jun 18, 2019

1) Subjective part are mostly from sets portion.
2) Only one question from truth table.
Jun 22, 2019

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

1) ek recursion ka tha
2) ek truth table ka tha
3) ek relation given tha
4) ek venn diagram ka tha (fill krna tha).
5) Ek long arithmetic sequence
6) dosra long Cartesian product ka tha (2 sets given thy).

Jun 23, 2018

1. Sor find krna tha ak set r dosra s ka given tha.
2. R^s is reflexive?
3. Gof(x) and fog (x)
4. P-> (p-~q) tatuolgy find krni the.
5. All subset of a= {1, 2, 3}?

Jun 10, 2017

1) Truth table bnana tha
2) set a ka relation r show krna tha whether it is symmetric or not...
3) Fog(x) k 2 questions thy or matrix sy show krna tha sor,

Dec 21, 2016

Question no 1. (Marks 2) If .Find fog(x).
Question no 2. (Marks 2) Write de Morgan's law for union and intersection.
Question no 3. (Marks 3) Show that the intersection of two transitive relations are also transitive.
Question no 4. (Marks 3) Prove by membership table.
Question no 5. (Marks 5) Identify the following logic laws.
Question no 6. (Marks 5) Find sequence n0, n1, n2, n3, n4 ... Using given formula

Dec 19, 2016

1) To generate a matrix from the given figure and also make a series 5
2) F: r implies z f (n) =under root (n) is a function or not? 5
3) Sequence to find -3 +1 -1 3 -5 ......a16? 2
4) Series given write range and domain? 2

Dec 23, 2015

1) Find domain and range 2 numb
2) Find sequence n0, n1, n2, n3, n4 ... Using given formula 5 numb
3) If g=f(x) =0, fn (fx) =2, 3b (fx)-3 find 2 (2numb)

Dec 26, 2015

Q) pair of sets given tha domain and range btani the.
Q) 2 matrices were given we have to find r.s and (r.s)t
Q) find the series whose second value is 9, fourth value is 1.
Q) if f(x)=x/2+3 and g(x)=3x/4-2, find the value of 5f(-2)-7g(-4).
Q) if f(x)=x3-1 and g(x)=x+2 , then find f.g(x)
Q) find the sum of series of first 15 natural numbers by using series formula.

Jun 28, 2015

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

Moaz ki file se 10 % objective aya tha baki relation se hi aye thy.
Subjective bi lect 15, 16, 17 se hi tha
One to one relation tha
Subject bi relation wale units se the.

Jan 21, 2015

1) f(0)=3, F(n+1)=4(fn)+1, Find f(3)?
2) Write the geometric sequence with positive terms whose second term is 9 and fourth term is 1.
3) Find (f-g) (x) where f(x) and g(x) were given
4) Show that x3 + 5 = y is onto
5) If set a has m elements and b has n elements. How many functions can exist from a to b?
6) Set x= {1, 2, 3,4} find a relation r on x {(x, y)r|x<y} also find r^{-1} and draw directed graph of r.

Jun 9, 2014

1) Show that the sequence 0, 1, 3, 7, . . . , 2 n – 1, . . . , for n ≥ 0, satisfies the recurrence relation dk = 3dk-1 – 2dk-2, for all integers k ≥ 2(5mark)
2) Find the first four arithmetic sequence of an=3n-5 (3mark)

Dec 27, 2013

1) ek question main domain range btani the
2) ek recursion waly lec sy function wala question.
3) Ek question ko prove krna tha k ye onto function hy k nae,
4) ek question arithmatic sequence sy 11th term find krni the 9th or 4th term btai hui the.
5) ek five marks ka recursion waly lec sy aya tha.

Dec 24, 2013

Q. 1 sawal unin or interscet ka aya tha esy tha ..
Q2. Gog(x) =? F(x)=2x + 1, g(x) = x^2+ 1
Q3. Let a {123} be set r and s be the following relations on its \
R = { (1,2) , (1,3) , (2,3) , (3,1) , (3,3) }
S = { (1,2) , (1,3) , (2,1) , (3,3) }
R ∩ s
Q4. Sum of the series x^2 + y^2 , x + y ....

Jun 2, 2013

1. R not s find for given matrix and also find transpose of it. 3 marks
2. Find a cross b for given sets. Is a cross b is equal to b cross a? 5 marks
3. Find domain and range for given set 2 marks
4. Sum the given arithmetic series to nth term. 5 marks
5. One question from recursion 2 marks

Dec 30, 2013

Q- lect-21, 2nd last example recursive--- wali ati thi.....
Q- values were give and i had to draw a directed graph of r.
Q- s and r were give and i needed to find the r intersection s.
Q- a set with 4 elements was given...i needed to find r for that with condition that a < b
Q- Also i needed to find r inverse for that

Q- And draw directed graph as well for this...

Jun 1, 2013

What is the difference between {a,b} and {{a,b}}?

What is the difference between yes and allowed in graphs?

Write the geometric sequence with positive terms whose second term is 9 and fourth term is 1?

what is the basic difference b/w sequences and series?

Let f and g be the functions from the set of integers to the set of integers defined by f(x) = 2 x + 3 and

g(x) = 3 x + 2 .What is the composition of f and g ? What is the composition of g and f?

Let R be the relation on from A to B as

R={(1,y) ,(2,x) ,(2,y),(3,x)}

Find

(a) domain of R

(b) range of R

May 25, 2013

1) Determine the matrix of the following?
2) Find the infinite g.p of 2,2^1/2,1?
3) Explain gof is one to one if and only if g(x) is one to one function?
4) F(x) =2x+3 and g(x)=3x-1 find fog and gof composition?
Let r be the relation from a to b asr={(1,1),(2,2),(3,3),(4,5),(4,4)}
5) Explain it is reflexive or not? and also find its ordered pair?
Let a ={ 1,2,3,4} find the binary relation of axa and r is {(1,1),(2,2),(3,3),(4,4),(2,1),(2,3)}
6) Find is it equivalent or not?

Jun 1, 2013

Q.no1 if f(x) = x and g(x)=-2x then (f+g)x = 3x–x
Q.no2 let the real valued functions f and g be defined by f(x) = 2x + 1 and g(x) = x2 – 1 obtain the expression for fog(x)
Q.no3 let a= {1, 5, 9} and b= {6, 7} then find Cartesian product from a to b and from b to a. Is both are equal or not also justify.

May 26, 2013

1) If relation r is transitive then shows that its inverse is also transitive???
2) Suppose that r and s are reflexive relations on a set a. Prove or disprove r,,xs is reflexive.
3) Find the 5th term of the g.p. 3,6,12,…
let f and g be the functions defined by f(x)= 2x+3 & g(x)= 3x+2 then find

(i) Composition of f and g. (ii) Composition of g and f.
4) lec 22 m say ek question aya tha
5) f(x) = x2 +1 and g(x) = x + 2 find (f.g)x and (f-g)x

May 19, 2012

Q # 1.determine f is function if f (n) =√n
Q # 2. How many term of series -9-6-3 up to 117
Q # 3. Let real valued functions f and g be defined by f(x)=2x+1 and g(x)=x2-1 obtain expression for fog(x).
Q # 4. Let r be the following relation on a={1,2,3,4}: R={(1,3) ,(1,4) ,(3,2), (3,3) ,(3,4)}
(a) Find the matrix m of r. (b) draw the directed graph of r. Also write its inverse
Q # 5. Find the matrix representing the relation sor and ros where matrix representing r and s are. Find matrix representing sor

May 11, 2012

Qs.No1: Find the 36^{th} term of the Arithmetic sequence if 3^{rd} term is 7 and 8^{th} term is 17. **5marks**

Qs.No2: If the function is f(x) = $\dfrac{2x+1}{2x+2}$ is it onto? **5marks**

Qs.No3: $if \quad a_n = \dfrac{1}{n^2+1}$ Find the 1^{st} 4 term of the series. **3marks**

Qs.No4: If f(x) = $3x^2+1$ and g(x) = x^2-1 then find gog(x) =?

Qs.No5: If set A= {1, 2, 3, 4} and B= {2, 3, 1}**,** R = { (1,1),(1,2),(2,2),(1,3),(3,3)**,** S= {(1, 1), (1, 3), (3, 3)} Find $R \cap S$ =?

Dec 17, 2012

1) Let a and b be integers. Suppose a function Q is defined recursively as follows:

$$Q(a,b) = \begin{cases} 2 & \text{if } a \langle b \\ Q(a+b, b-1) + a & \text{if } b \le a \end{cases}$$

Find the value of Q (5, 2)

2) Let a and b be integers. Suppose a function Q is defined recursively as follows:

$$Q(a,b) = \begin{cases} 5 & \text{if } a \langle b \\ Q(a+b, b-1) + a & \text{if } b \le a \end{cases}$$

Find the value of Q (14, 5)

3) 1 qs is from sequence lecture # 19, Find the … term of geometric sequence Kuch is tra ke he tha qs

4) 1 qs me show krna tha k function symmetric reflexive or transitive hay? Function simple he tha.

Dec 17, 2012

1) Find the domain and range r= {(1,x)(2,y)(2,x)(3,y), sum the series -3+(-1)+1+3+5………..
2) Draw the directed graph if, a={1,2,3,4,6},r={(1,1)(1,3)(1,6)(1,2)(1,4)(2,2)(4,4)(6,6)(3,3)(2,4)}

3) if f(0)=3, f(n+1)=4f(n+1), then find the value of f(3)
4) find the geometric series if its second term is 9 and 4th term is 1.

Nov 30, 2011

1) 1 short q is about range and domain. 2 marks
2) 1 q is find anti-symmetric and transition of given r. 2 marks
3) 1 is series find and an=59,d=7,a=3. 5 marks
4) which is the term of the series{(1*4),(2*7),(3*10).........} 5 marks.
5) 1 is cartesian product of a and b set and a*b and b*a find and give your view that is equal. a={1,3,5,10},b={1,6,9,7}
6) 1 is about* find g.p* and data nahi pata

Nov 30, 2011

1) given a={1,2,3,4} and b = {x,y,z} let r be the following relation for a to b r = {(1,y),(1,z),(3,y),(4,x),(4,z)} determine the matrix of the relation. 2 number
2) suppose that f is recurisvely by f(0) = 3, f(n+1) = 2 f(n) + 3, find f(1)? 2 number
3) let the real valued functions f and g be defined by f(x) = 2x + 1 and g(x) =- x^2 + 1 obtain the expression for g^2(x) 3 number
4) given a = {1,2,3,4} consider the relation with a r = {(1,1),(2,2),(2,3),(3,2),(3,3),(4,2),(4,4) is r (a) transitive (b) antisymmetric justify your answer 3 number
5) consider the following relation on the set a={1,2,3} r = empty set determine whether or not the given relation on "a" is (i) reflexive. (ii) transitive. (iii) antisymmetri (justify the answer)
Q no. 6 write the geometric sequence with positive terms whose second term is 9 and fourth term is 1. 5 number

Dec 10, 2010